

Top- $(K+J)$ with the RBLapSum surrogate versus hard Top- K'

Research note: what the candidate window buys

2026-09-22.

1 Purpose and summary of conclusions

The RBLapSum `through_rank_kappa` bottleneck keeps K features in the forward pass and gives gradient to a candidate pool of $K + J$. A hard Top- K' bottleneck with $K' = K + J$ keeps the whole pool. The two therefore see the same number of candidate features, but the surrogate model activates fewer of them: J enters only the backward pass, and inference skips the surrogate machinery entirely.

This note asks what the candidate window is worth. Two questions are separable:

- at a fixed *candidate pool* $K' = K + J$, does the surrogate with K active features reach the loss of hard Top- K' , which activates all K' ?
- at a fixed *active count* K , how much does widening J help?

Both variants identified in the stabilization note are included, since the better one was configuration-dependent there: $T = 2$, and an RMSNorm on each bottleneck's output at $T = 1$ ("post-norm"). All hard baselines carry the post-norm as well.

The main conclusions are:

- **With the post-norm, a pool twice the active count beats hard Top- K' outright, at every scale tested.** For $J = K$ – that is, $K' = 2K$ – the surrogate model wins in all four pools, by 0.009 to 0.014 nats, while activating half as many features: 1.5310 vs 1.5408 at $K'=64$, 1.4834 vs 1.4938 at $K'=128$, 1.4423 vs 1.4560 at $K'=256$, and 1.4146 vs 1.4232 at $K'=512$.
- **The best configuration in the comparison is $K = 256$, $J = 256$ with the post-norm,** at 1.4146: better than hard Top-512 (1.4232) with half the active features, and better than hard Top-256 (1.4560) by 0.041 nats at the same active count.
- **At a matched active count the surrogate is worth 0.04–0.11 nats**, and the margin shrinks as K grows: 0.107 at $K = 32$, 0.087 at $K = 64$, 0.064 at $K = 128$, 0.041 at $K = 256$.
- **Widening J helps monotonically, except at the extreme.** At $K = 64$, 128 and 256 every increase in J improves the loss. At $K = 32$ the improvement stops and reverses at $J = 480$: post-norm goes $1.5310 \rightarrow 1.5115 \rightarrow 1.4949$ for $J = 32, 96, 224$ and then back to 1.5370 at $J = 480$.
- $T = 2$ **never reaches hard Top- K'** , trailing the matched hard baseline by 0.008–0.024 nats in all four pools, and it is worse than the post-norm in 8 of the 10 cells. The two exceptions are both at $K = 32$ with the widest windows ($J = 224$ and $J = 480$), which is also where the post-norm is weakest – the same configuration dependence the stabilization note found.

- **No run diverged**, including both $K = 256$ cells, which the initialization screening had flagged as the most likely to fail.

2 Setup

Twenty-five cells, all under full magnitude–direction decoupling with the architecture, data, optimizer, schedules and seed of the earlier campaigns (8×768 , `abs_topk` over $N = 1536$ features, `residual_out` placement, batch 96×512 tokens, 20k steps, $b_0 = 0$).

The grid is $K, K' \in \{32, 64, 128, 256, 512\}$ with $J = K' - K > 0$, which gives ten (K, K') pairs; $K' = 32$ has no partner because the K grid starts there. Each pair was trained under both variants, and each K' has a hard Top- K' baseline with the post-norm:

arm	kernel / normalization	active features
$T = 2$	$T = 2$, no output norm	K
post-norm	$T = 1$, RMSNorm per bottleneck output	K
hard Top- K'	no surrogate, RMSNorm per bottleneck output	K'

Eleven runs were trained for this note: the $K = 256$, $J = 256$ cell in both variants; the post-norm cells at $(32, 96)$, $(32, 224)$ and $(64, 192)$; the same three cells re-run at $T = 2$ with $b_0 = 0$, replacing older $b_0 = 0.1$ runs; and hard Top- K' with post-norm at $K' = 64, 128$ and 256 . The remaining fourteen cells are the matching arms of the stabilization campaign and the two hard post-norm runs from the preceding note. Every cell used in the tables below is therefore a $b_0 = 0$ run; where a cell had been trained more than once, the selection order is stabilization campaign, then this campaign, then the kappa ablation, and `scripts/plot_kj_comparison.py` prints the run backing each cell.

One asymmetry is worth keeping in view when reading the tables. The candidate window costs nothing at inference: `hard_inference` skips the surrogate, so a trained model’s forward pass depends on K and not on J . Within the bottleneck the encoder projection is dense in N regardless, while the decoder reads a K -sparse code, so the K -dependent part of the cost is roughly proportional to K . A cell that matches hard Top- K' at $K < K'$ is thus the cheaper model at equal quality.

3 Results

Final validation CE after 20k steps. Rows are the active count K , columns the candidate pool $K' = K + J$, and the last row is hard Top- K' with the post-norm. Bold marks the cells that beat the hard baseline in their own column.

$T = 2$				
K	$K'=64$	$K'=128$	$K'=256$	$K'=512$
32	1.5492	1.5146	1.4842	1.4943
64	—	1.5142	1.4922	1.4727
128	—	—	1.4802	1.4598
256	—	—	—	1.4425
hard Top- K'	1.5408	1.4938	1.4560	1.4232

post-norm

K	$K'=64$	$K'=128$	$K'=256$	$K'=512$
32	1.5310	1.5115	1.4949	1.5370
64	—	1.4834	1.4602	1.4538
128	—	—	1.4423	1.4294
256	—	—	—	1.4146
hard Top- K'	1.5408	1.4938	1.4560	1.4232

The bold entries are exactly the diagonal $J = K$. Under the post-norm every pool is won by its $K = K'/2$ cell and by no other: at $K' = 512$, for instance, $K = 256$ wins by 0.0086, $K = 128$ trails by 0.0062, $K = 64$ by 0.031 and $K = 32$ by 0.114. Under $T = 2$ no cell wins any pool.

At a matched active count. Read along a row instead, against the hard baseline with the *same* active count:

K	best post-norm cell	hard Top- K	difference
32	1.4949 ($J = 224$)	1.6022	−0.1073
64	1.4538 ($J = 448$)	1.5408	−0.0870
128	1.4294 ($J = 384$)	1.4938	−0.0644
256	1.4146 ($J = 256$)	1.4560	−0.0414

The surrogate is worth between 0.04 and 0.11 nats at fixed activation cost, and the margin narrows steadily with K – consistent with the candidate window mattering most when the forward support is most restrictive.

The effect of widening J . At fixed K , with the post-norm:

K	final validation CE by J				trend
32	$J=32$: 1.5310	$J=96$: 1.5115	$J=224$: 1.4949	$J=480$: 1.5370	reverses
64	$J=64$: 1.4834	$J=192$: 1.4602	$J=448$: 1.4538		monotone
128	$J=128$: 1.4423	$J=384$: 1.4294			monotone
256	$J=256$: 1.4146				—

The reversal at $K = 32$, $J = 480$ is a window fifteen times the active count, and it is the one cell where $T = 2$ clearly beats the post-norm (1.4943 against 1.5370). Both facts point the same way: a very wide window relative to K is the regime where the surrogate’s boundary term is hardest to control, and where the kernel width rather than the output norm is the effective remedy.

Per candidate pool

Each figure fixes the pool $K' = K + J$ and compares hard Top- K' (dashed, black) with every surrogate cell sharing that pool (solid, a blue ramp running light to dark with increasing J). Left column $T = 2$, right column post-norm; the lower row repeats both panels with the vertical axis capped at 2.0 nats.

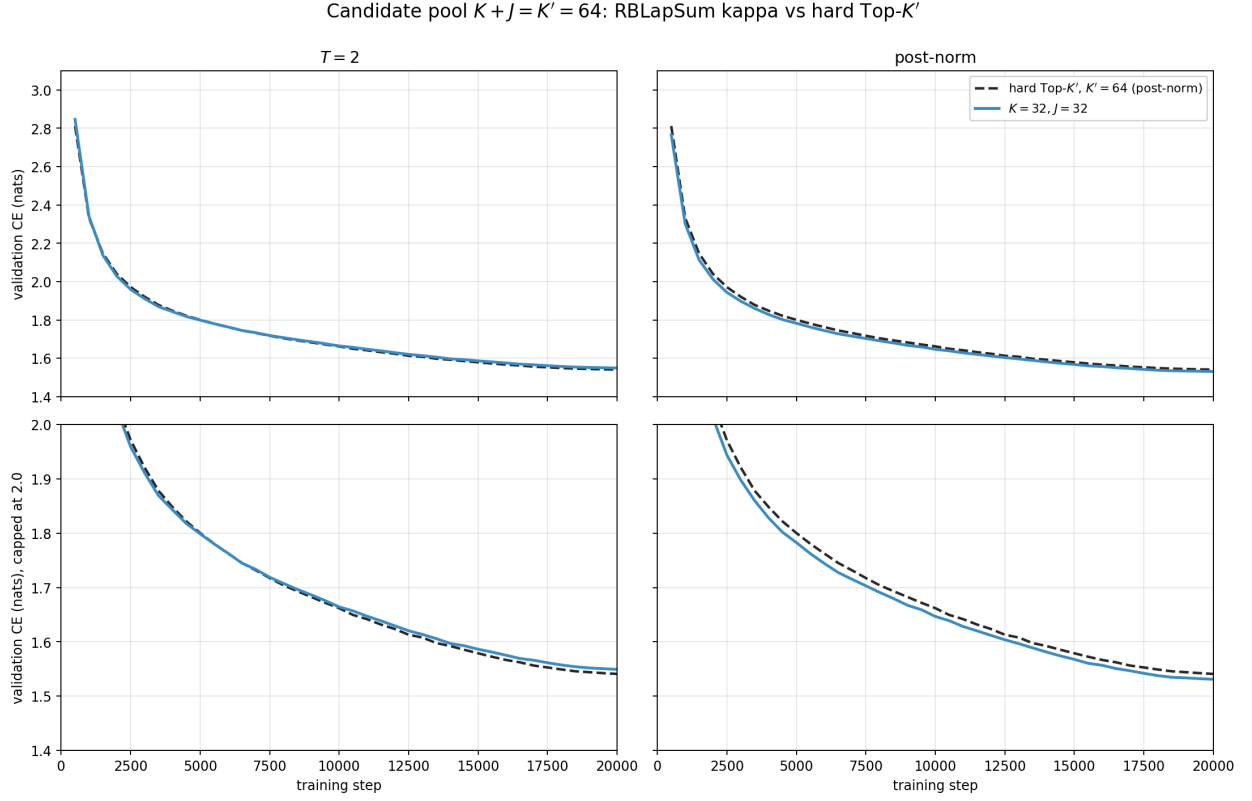


Figure 1: $K' = 64$. The single surrogate cell ($K = 32, J = 32$) ends below hard Top-64 under the post-norm and slightly above it at $T = 2$.

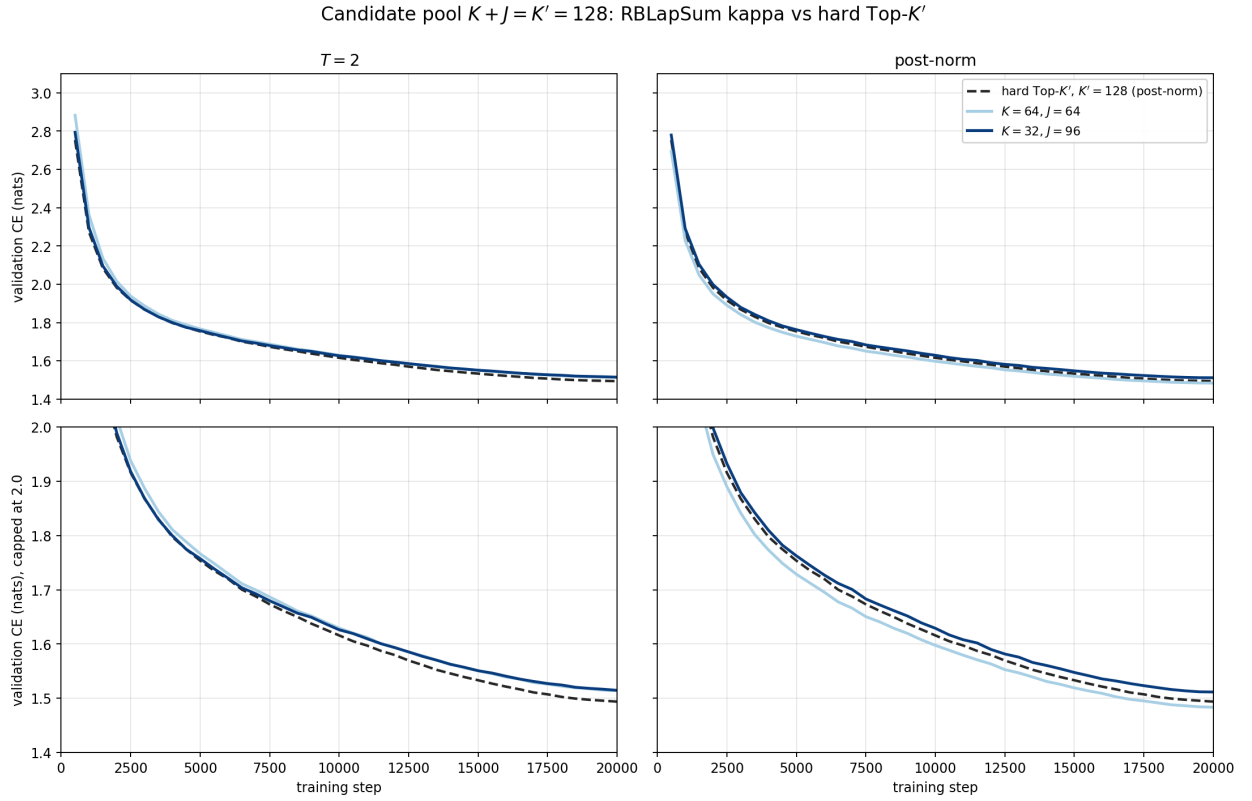


Figure 2: $K' = 128$. $K = 64$ wins the pool under the post-norm; $K = 32$, with the same pool but half the active count, trails the hard baseline.

Candidate pool $K + J = K' = 256$: RBLapSum kappa vs hard Top- K'

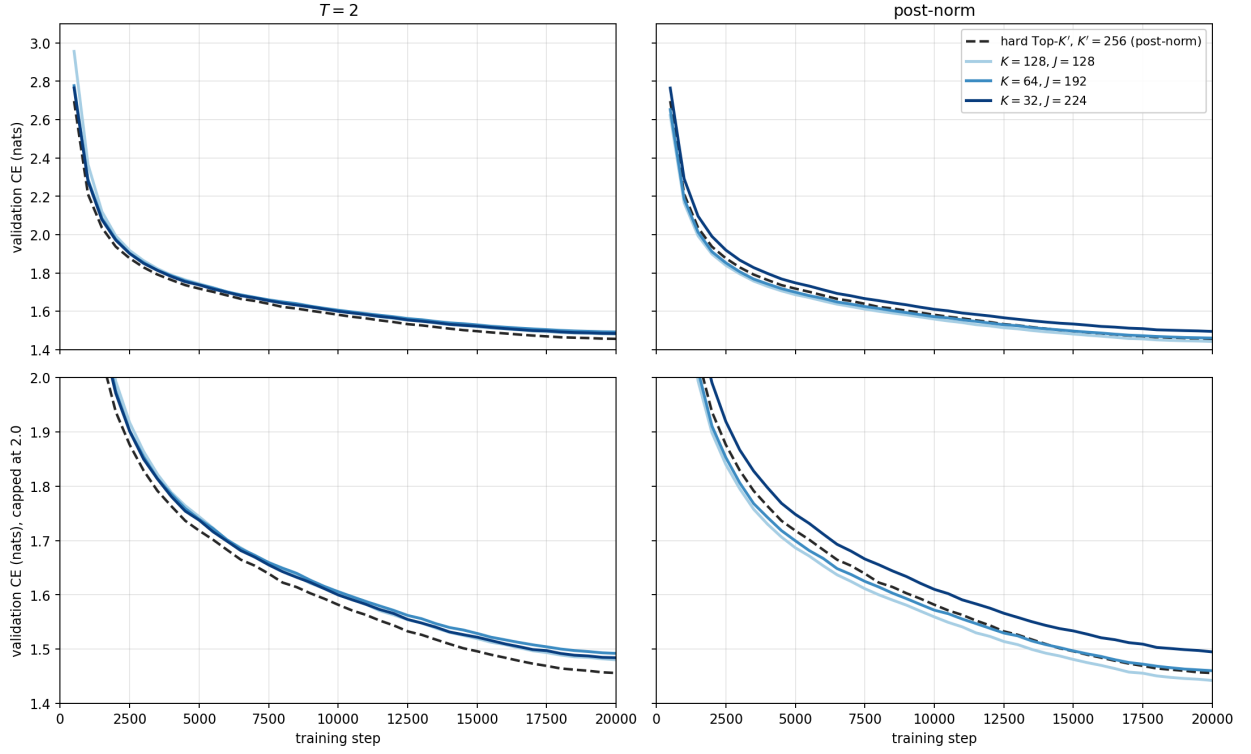


Figure 3: $K' = 256$. Under the post-norm the $K = 128$ curve separates from the hard baseline from about step 5000 and stays below it; $K = 64$ converges to within 0.004 nats of hard, and $K = 32$ remains clearly above. At $T = 2$ all three sit above the baseline throughout.

Candidate pool $K + J = K' = 512$: RBLapSum kappa vs hard Top- K'

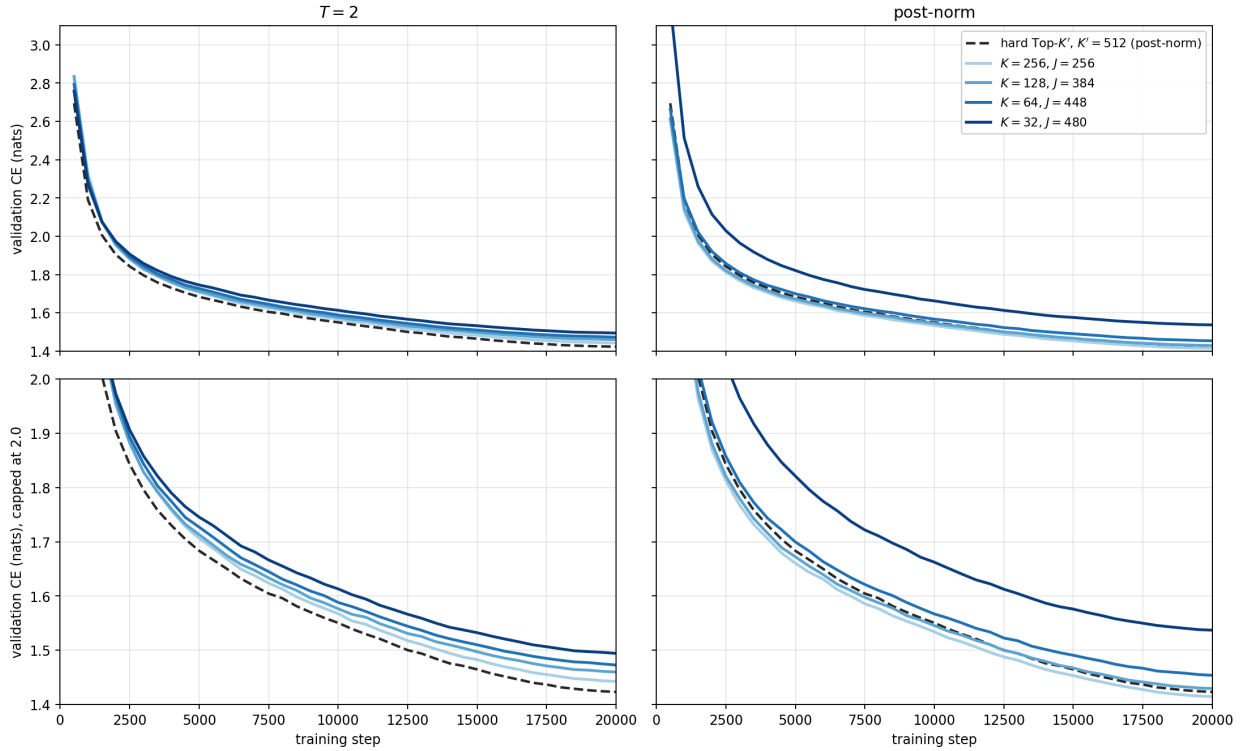


Figure 4: $K' = 512$. The ordering is monotone in K : $K = 256$ ends below hard Top-512, $K = 128$ tracks it to within 0.006 nats, and $K = 64$ and $K = 32$ fall progressively behind. The $K = 32$ curve is the reversal discussed above.

Per active count

Each figure fixes the active count K and shows every candidate window trained at that K (solid, a blue ramp light to dark with increasing J) against all five hard Top- K' baselines (dashed, a red ramp light to dark with increasing K'). Hue separates the two families and lightness orders each, so a surrogate curve is never confusable with a hard one. The hard curves are identical in all four figures, so they serve as a fixed reference scale.

Active count $K = 32$: every candidate window, against all hard Top- K'

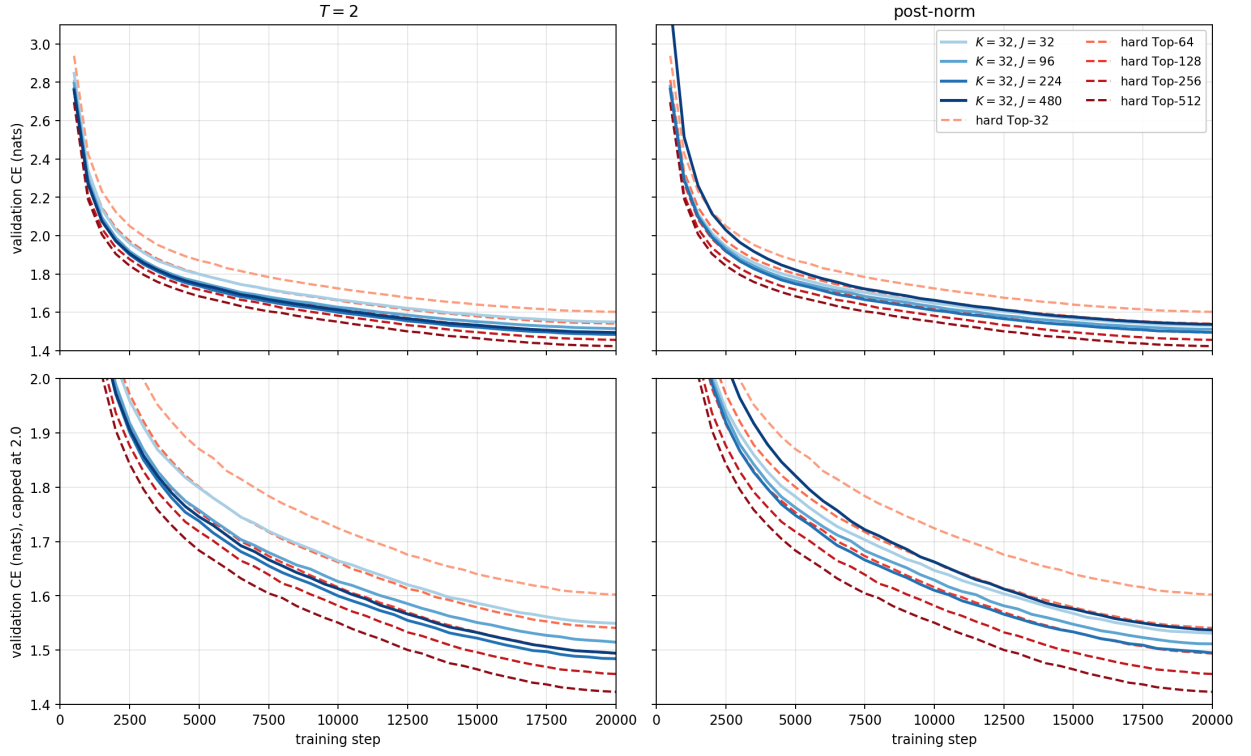


Figure 5: $K = 32$. Widening the window moves the model up the hard ladder – from between hard Top-32 and Top-64 at $J = 32$ to past hard Top-128 at $J = 224$ – but the widest window, $J = 480$, falls back under the post-norm while continuing to improve at $T = 2$.

Active count $K = 64$: every candidate window, against all hard Top- K'

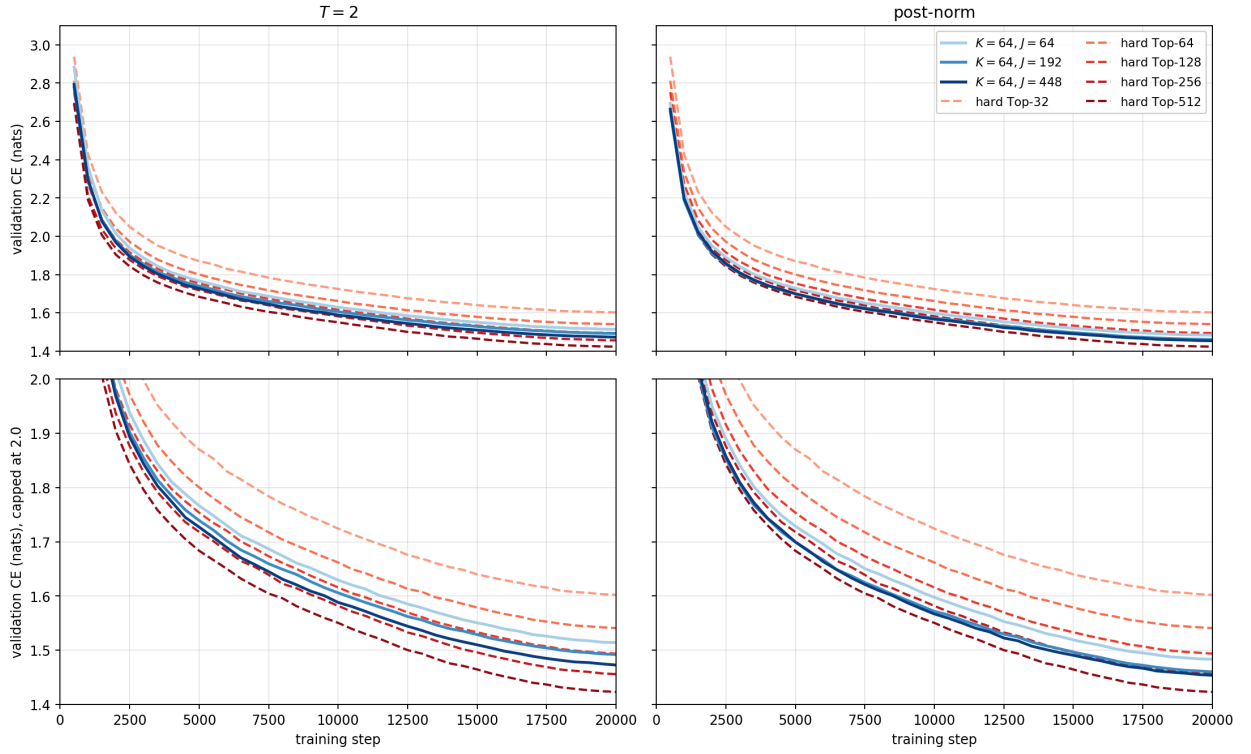


Figure 6: $K = 64$. Every window improves on the last, and the best ($J = 448$) lands between hard Top-256 and hard Top-512 while activating 64 features.

Active count $K = 128$: every candidate window, against all hard Top- K'

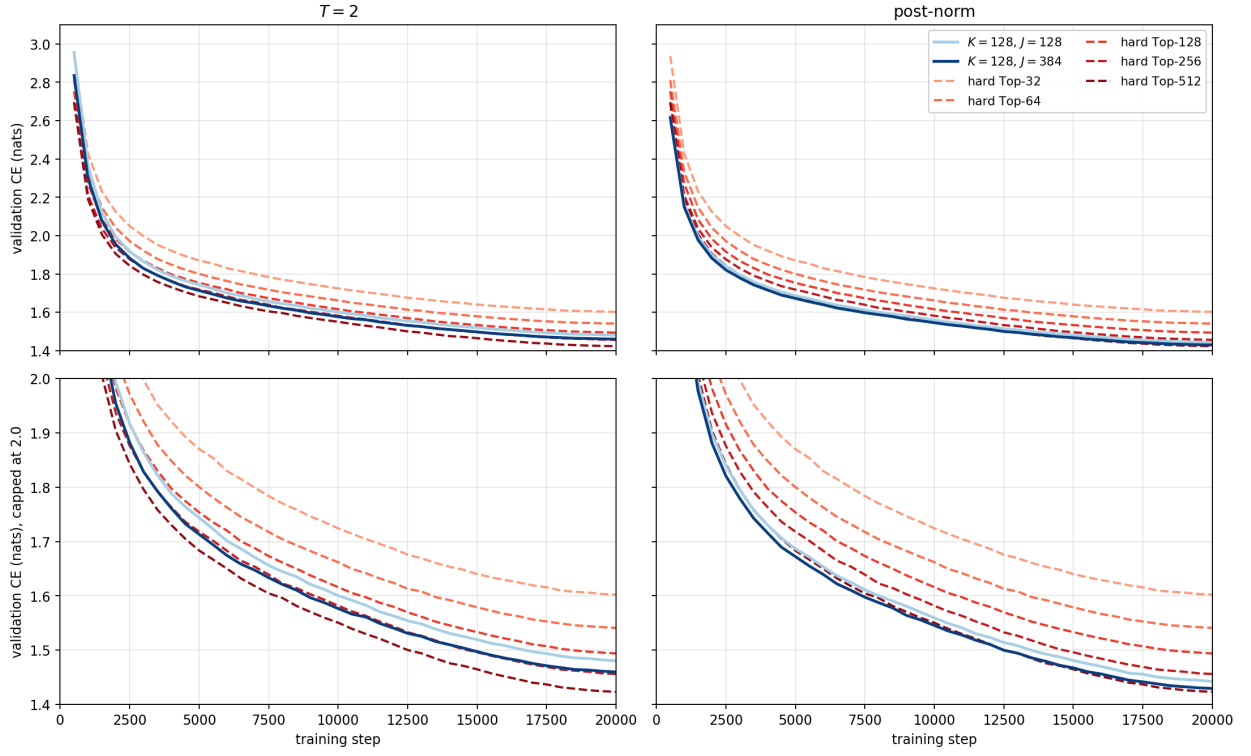


Figure 7: $K = 128$. Both windows beat hard Top-256 under the post-norm, and $J = 384$ reaches hard Top-512 to within 0.006 nats at a quarter of its active count.

Active count $K = 256$: every candidate window, against all hard Top- K'

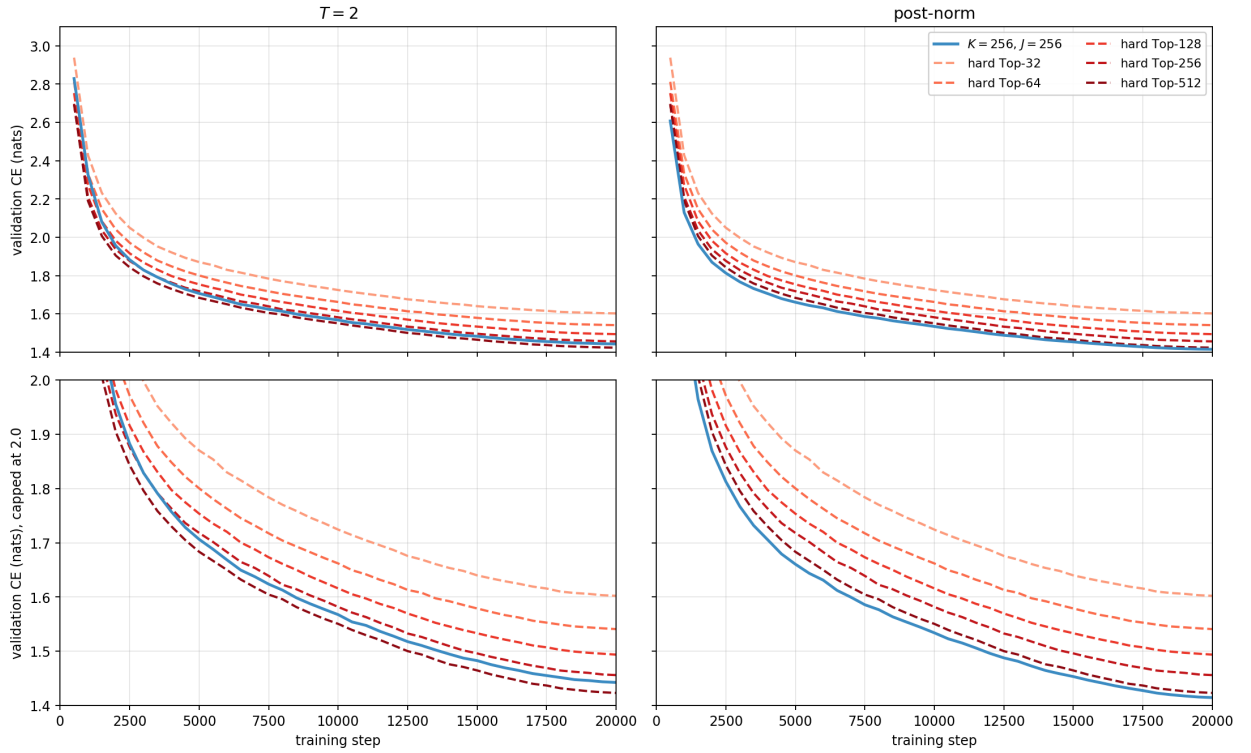


Figure 8: $K = 256$. Under the post-norm the single cell ($J = 256$) ends below every hard baseline including Top-512, which is the best result in the comparison. At $T = 2$ the same cell sits between hard Top-256 and Top-512.

4 Interpretation

Two regularities organize the tables.

First, **the candidate window substitutes for active features, at a roughly fixed exchange rate of about a factor of two.** Under the post-norm, K active features with a pool of $2K$ beat $2K$ active features with no surrogate, and they do so at all four scales tested, with a margin (0.009–0.014 nats) that is remarkably stable across a factor of eight in K . Pushing further – a pool of $4K$ or more – keeps improving the loss at fixed K but no longer beats the correspondingly larger hard model: at $K' = 512$, $K = 128$ (pool $4K$) lands 0.006 nats short of hard Top-512 and $K = 64$ (pool $8K$) 0.031 short. The surrogate closes roughly one doubling of the active count, not an unbounded amount.

Second, **the post-norm is the better variant almost everywhere, and the exceptions need both a small active count and a very wide window.** It wins 8 of the 10 cells, by 0.003–0.038 nats; the two losses are $K = 32$ with $J = 224$ and $J = 480$, where $T = 2$ is ahead by 0.011 and 0.043. The ratio alone does not explain it: $K = 64$ with $J = 448$ has the same $J/K = 7$ as the first loss and still favours the post-norm by 0.019. What the two exceptions share is a small support inside a window seven to fifteen times wider – the regime where the kernel must span far more candidates than the model keeps, and where widening the kernel is the effective remedy while normalizing the output is not. This is a sharpening of the stabilization note’s observation that the better variant was configuration-dependent, but it remains an observation about two cells and not a mechanism.

Together the two regularities suggest a practical default: a pool of about $2K$ with the output norm, and $T = 2$ instead if the window must be much wider than the support.

5 Scope

All cells are single runs at one seed. The differences that carry the conclusions are of two sizes: the matched-active-count margins (0.04–0.11 nats) are large relative to anything seen from run-to-run variation in this project, while the diagonal wins over hard Top- K' (0.009–0.014 nats) are small, and rest on their consistency across four independent pools rather than on any single comparison. A seed repeat of the four diagonal cells and their hard baselines would settle that directly.

The hard baselines all carry the post-norm, which the previous note measured as a small improvement for hard bottlenecks (−0.014 at $K = 32$, −0.007 at $K = 512$); against hard baselines without it the surrogate’s margins would be slightly larger. Only two variants were tested, and only one value of each ($T = 2$; RMSNorm at $T = 1$). Finally, $K + J \leq 512$ throughout, well inside $N = 1536$, so nothing here speaks to pools approaching the full feature set.